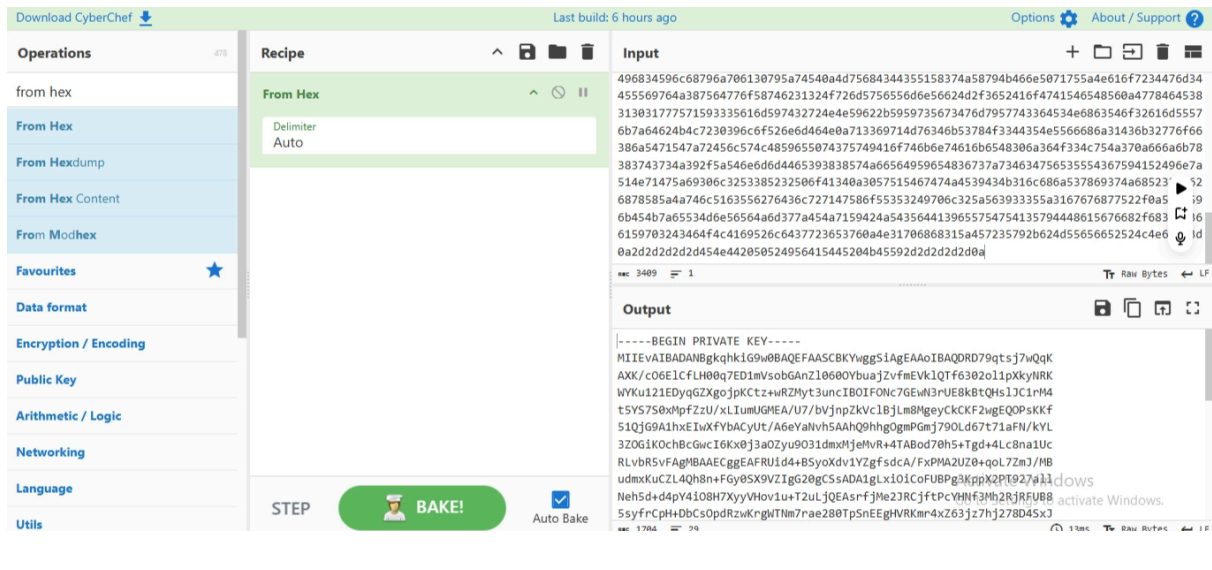


1



Step 3: Reconstructing the Private Key File

The decoded key was saved into a `.pem` file for use with OpenSSL.

```
echo "-----BEGIN PRIVATE KEY-----
MIIEvAIBADANBgkqhkiG9w0BAQEFAASCbKYwggSiAgEAAoIBAQRDRD79qtsj7wQqK
AXK/c06ElCfLH00q7ED1mVsobGAnZl0600YbuaJZvfmEVklQTf6302o1lPxkyNRK
WYKu121EDyqGZXgojpKctz+wRZMyt3uncIB0IFONc7GEwN3rUE8kbtQHsLJC1rM4
t5YS7S0xMpfZzU/xLIumUGMEA/U7/bVjnpZkVclBjLm8MgeyCkCKF2wgEQ0PsKKf
51QjG9A1hxEIwXfYbACyUt/A6eYaNvh5AAhQ9hhgOgmPGmj790Ld67t71aFN/kYL
3ZOGiK0chBcGwcI6Kx0j3a0Zyu9031dmxMjEmVr+4TABod70h5+Tgd+4Lc8na1Uc
RLvbR5vFagMBAAECggEAFRUId4+BSyoXdv1YZgfsdcA/FxPMA2UZ0+qoL7ZmJ/MB
udmxKuCZL4Qh8n+FGy0SX9VZIgG20gCSsADA1gLxi0iCoFUBPg3KppX2PT927a1l
Neh5d+d4pY4i08H7XyyVHov1u+T2uLjQEAsrfjMe2JRCjftPcYHNf3Mh2RjRFUB8
5syfrCpH+DbCs0pdRzwKrgWTNm7rae280TpSnEEgHVRKmr4xZ63jz7hj278D4SxJ
qiB1CiU13LRC5Y9Pry/qnwfNAf8D1xyiokDTsKI/xoosTU6QIXyKel0P9rB3WjvS
GZVSTgAayMp6LNQKf106tQo1UsF/c1S3fArbVnoXQQKBgQDf0dtrhn1vXkHVh1ef
tTokCMUfq/BDKDC37pedFQXr/8HeWjoi6P7L00SjbHcR7l/nItEPA8pHpCJjnBgb
c28rGP/6X0pxUyL7pG4IVl90sSwpeeDenazMfufFG5HC0wVRp+N85oBi+VNe15yu
fBI427uuGpaqEx7e02GuQeljEQKBgQDvwYGIMAEEiQzhWkThoZRYsm17SLNojx6p
sLAlvMplwj8jdSGRrUF0p1H5s0bhBiT4YA5kJexAN7DwNhC0LqI7+VoR/4e/D/Lj
4G0bCERghJ694zlhuxFuoQb6MsAz2uaZMcB+FAkmTp4WS1zfdbgPv0rFFNLnsNv7
iuq/Jc4FdQKBgA1GCncdWy4ZI78Huj0j8J4d62LTvYiwxhZz0igkD0MYpTWmLfJn
emcyN7ZEB9yIGSnMVzQWXXJvf9a+Io6JWPeEJLod7zRhqAiWGHih4Ylhyjpa0yZtT
MuhCD5QX7JXyKFnPquZNaor4Gm4EUivJ8udwoXtb120rmWVUmnVbM/6RAoGATeHV
GxFE8101wuqY35amYt2rNNYb+YYsVsGmyWt3dSNhcTo2amUWkzdbKLr09loRnmFN
q3iqMv4kSx03D5NUfhj1Ck2wof8jTqTzrElWLHYePt7WIAotkntakeH0j603LuJ7
fjKx87CsJ9/ZTnmDe988WJfVIYeH6szsF4ue5UCgYARInzQNqGZi0l2S8R2PoA4
0WQTgGJE9CK1lhjSxi7JhR22bhxXZJtlQcUbvClrqGXoU52Ip12ZV935Z1gghwR/
```

```
QiYkEKzeSMnVVJm7zEJqYBJT5dA9eWTuA5yDHaVvh/h3i6aYp2CF0LAiRld7r6Sv
Nlphh1ZEr5y+bMUefRRLNg==
-----END PRIVATE KEY-----
" > privkey.pem
```

Step 4: Decrypting the Flag

Once the private key was reconstructed, OpenSSL was used to decrypt the encrypted flag file.

A terminal window with a dark purple background. The prompt is 'loki@SolarisFortress: ~/Desktop'. The command entered is 'openssl rsautl -decrypt -inkey privkey.pem -in flagRSA.enc'. A message appears: 'The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead. picoCTF{rs4_k3y_1n_1mg_0a64c2f9}'. The prompt changes to 'loki@SolarisFortress: ~/Desktop\$'. Below the terminal, a file explorer bar shows 'flagRSA.enc' with a size of '256 bytes' and a status of 'Unknown'. On the right side of the terminal window, there is a 'Activate Windows' watermark with the text 'Go to Settings to activate Windows.'

```
openssl rsautl -decrypt -inkey privkey.pem -in flagRSA.enc
```

Result

```
picoCTF{rs4_k3y_1n_1mg_0a64c2f9}
```